



Review

Coffee and caffeine intake and breast cancer risk: An updated dose–response meta-analysis of 37 published studies

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HIGHLIGHTS

- Coffee and caffeine might be weakly associated with breast cancer risk for postmenopausal women.
- A strong and significant association was found for BRCA1 mutation carriers.

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ABSTRACT

Objective. We conducted an updated meta-analysis to summarize the evidence from published studies regarding the association of coffee and caffeine intake with breast cancer risk.

Methods. Pertinent studies were identified by a search of PubMed and by reviewing the reference lists of retrieved articles. The fixed or random effect model was used based on heterogeneity test. The dose–response relationship was assessed by restricted cubic spline model and multivariate random-effect meta-regression.

Results. 37 published articles, involving 59,018 breast cancer cases and 966,263 participants, were included in the meta-analysis. No significant association was found between breast cancer risk and coffee ($RR = 0.97$, $P = 0.09$), decaffeinated coffee ($RR = 0.98$, $P = 0.55$) and caffeine ($RR = 0.99$, $P = 0.73$), respectively. And the association was still not significant when combining coffee and caffeine (coffee/caffeine) ($RR = 0.97$, $P = 0.09$). However, an inverse association of coffee/caffeine with breast cancer risk was found for postmenopausal women ($RR = 0.94$, $P = 0.02$), and a strong and significant association of coffee with breast cancer risk was found for BRCA1 mutation carriers ($RR = 0.69$, $P < 0.01$). A linear dose–response relationship was found for breast cancer risk with coffee and caffeine, and the risk of breast cancer decreased by 2% ($P = 0.05$) for every 2 cups/day increment in coffee intake, and 1% ($P = 0.52$) for every 200 mg/day increment in caffeine intake, respectively.

Conclusions. Findings from this meta-analysis suggested that coffee/caffeine might be weakly associated with breast cancer risk for postmenopausal women, and the association for BRCA1 mutation carriers deserves further investigation.

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Contents

Introduction	621
Methods	621
Literature search and selection	621
Data extraction	621
Statistical analysis	621
Results	622
Literature search and study characteristics	622
Quantitative synthesis	622
Overall association of coffee and caffeine with breast cancer risk	622
Subgroup analysis by study design	622

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Subgroup analysis by population subgroups	622
Subgroup analysis by adjustment for covariates	622
Dose–response analysis	622
Association of coffee intake with breast cancer	622
Association of caffeine intake with breast cancer	625
Sensitivity analysis and publication bias	625
Discussion	625
Conflict of interest statement	628
References	628

Introduction

Breast cancer is the one of the most frequently diagnosed cancer in women, and ranks second as a cause of cancer death in women (after lung cancer) [1]. An estimated 226,870 new cases of invasive breast cancer, 63,300 new cases of in situ breast cancer and 39,510 breast cancer deaths are expected among women in the US during 2012, and the breast cancer incidence rates are stable since 2004 [1]. Coffee is one of the most popular beverages in the world, and the latest coffee trade statistics estimated that world coffee export amounted to about 6.76 billion kg in 2011/2012 [2]. The association between coffee intake and breast cancer risk has been investigated since the early 1970s [3], and many epidemiologic studies have been published on coffee or caffeine intake and breast cancer risk. However, according to the World Cancer Research Fund/American Institute for Cancer Research in 2008, the result was still inconclusive on coffee intake and breast cancer risk for both premenopausal and postmenopausal women [4]. A meta-analysis is available on coffee intake with breast cancer risk [5]. 10 studies (8 cohort studies [6–13] and 2 case–control studies [14,15]) were published thereafter, and we additionally identified 10 studies (1 cohort [16] and 9 case–control studies [17–25]) that were published before the meta-analysis. The association of caffeine intake with breast cancer risk is not summarized, and the association of coffee intake with breast cancer risk by menopausal status, body mass index (BMI), estrogen receptor (ER) and progesterone receptor (PR) status, breast cancer stage, and adjustment for important clinical and lifestyle factors is still unknown. Besides, the dose–response relationship, which is essential for proving causality, is also unknown. In addition, categories of coffee and caffeine intake levels differed between studies, which might complicate the interpretation of the pooled results across study populations with different categories. In this respect, a dose–response meta-analysis with restricted cubic spline functions provides a solution to the problem. Therefore, we conducted an updated dose–response meta-analysis to explore the above-mentioned issues in this study.

Methods

Literature search and selection

We performed a literature search up to Dec 2012 using PubMed database with the key words coffee or caffeine combined with breast cancer without restrictions. Furthermore, the reference lists of retrieved articles were scrutinized to identify additional relevant studies.

Two investigators independently reviewed the identified studies, and studies were included if they met the following criteria: (1) the study was conducted in humans; (2) the exposure of interest was coffee or caffeine; (3) the outcome of interest was breast cancer; and (4) relative risk (RR) with 95% confidence interval (CI) was provided (we presented all results with RR for simplicity). For dose–response analysis, the study had to report RR (95% CI) for at least 3 quantitative categories of coffee or caffeine intake. Besides, the number of cases and participants or person-years for each category of coffee or caffeine intake must be also provided (or data

available to calculate them). If data were duplicated in more than one study, we included the study with the largest number of cases.

Data extraction

The following data were extracted from each study by two investigators: the first author's last name, publication year, the name of the cohort study, country where the study was conducted, years of follow-up, age, sample size (number of cases and total number of participants), coffee and caffeine intake categories, covariates adjusted for in the multivariable analysis, and the RRs with their 95% CIs for each category of coffee and caffeine intake. We extracted the RRs that reflected the greatest degree of adjustment for potential confounders. For dose–response analysis, the number of cases and participants (person-years) for each category was also extracted. The median or mean level of coffee and caffeine intake for each category was assigned to the corresponding RR for every study. If the upper boundary of the highest category was not provided, we assumed that the boundary had the same amplitude as the adjacent category [26].

Statistical analysis

Pooled measure was calculated as the inverse variance-weighted mean of the logarithm of RR (95% CI) of breast cancer for the highest versus lowest category of coffee and caffeine, respectively. The I^2 of Higgins and Thompson was used to assess heterogeneity [27]. I^2 is the proportion of total variation contributed by between-study variation, and I^2 values of 0, 25, 50, and 75% represent no, low, moderate, and high heterogeneity [27], respectively. If moderate or lower heterogeneity ($I^2 < 50\%$) was found, the fixed effect model (FEM) was used as the pooling method, otherwise, the random effect model (REM) was adopted ($I^2 > 50\%$) that considers both within-study and between-study variations. A sensitivity analysis was performed with one study removed at a time to assess whether the results could have been affected markedly by a single study. Publication bias was evaluated using the Egger regression asymmetry test. Subgroup analysis was performed by study design (cohort study or case–control study), follow-up duration for cohort study (< 10 years or > 10 years), source of controls for case–control study (population-based or hospital-based), menopausal status (premenopausal or postmenopausal), ER and PR status (ER+PR+, ER+PR–, ER–PR+ or ER–PR–), BMI (< 25 kg/m² or > 25 kg/m²), breast cancer stage (in situ or invasive) and country where the study was conducted (USA, Europe or Asia). Besides, subgroup analysis was also performed by adjustment (yes or no) for smoking and/or alcohol, BMI, total energy intake, physical activity, oral contraceptive use, postmenopausal hormone replacement therapy use, family history of breast cancer and history of benign breast disease.

For dose–response analysis, a two-stage random-effects dose–response meta-analysis [28] was performed to compute the trend from the correlated log RR estimates across levels of coffee and caffeine, respectively, taking into account the between-study heterogeneity. In the first stage, a restricted cubic spline model with three knots at the 25th, 50th, and 75th percentiles [29] of the levels of coffee and caffeine, was estimated using generalized least square regression taking

into account the correlation within each set of published RRs. Then the study-specific estimates were combined using the restricted maximum likelihood method in a multivariate random-effects meta-analysis [30]. A *P* value for nonlinearity was calculated by testing the null hypothesis that the coefficient of the second spline is equal to 0 [29]. All statistical analyses were performed with STATA version 12.0 (Stata Corporation, College Station, TX, USA). All reported probabilities (*P* values) were two-sided with $P \leq 0.05$ considered statistically significant.

Results

Literature search and study characteristics

The search strategy identified 239 articles, of which 195 articles were excluded after review of the title or abstract (Fig. 1). 44 full-text articles were reviewed. We further excluded 4 articles that did not provide the independent result on coffee and breast cancer, and 3 articles were also excluded because RR and/or 95%CI were not provided. The remaining 37 studies [6–25,31–47] were included in this meta-analysis. 1 study (providing the result for coffee) [7] was a duplicate report from the same population of a previous study (providing the result for coffee and caffeine) [40], thus we included the result for coffee from the recent study [7] because of larger number of cases included. The detailed characteristics of the 37 studies are shown in Table 1.

Among the 37 studies, 28 studies provided the result for coffee [7–9,11,13–23,25,31,32,34–36,39,41–44,46,47], and 6 studies [6,10,12,24,37,45] provided the result for coffee and caffeine separately, and 3 studies [33,38,40] provided the result for caffeine. Because the association of breast cancer risk with coffee was similar with that of breast cancer risk with caffeine in the 6 studies, and the combined result of breast cancer risk with coffee was also similar with that of breast cancer risk with caffeine in this meta-analysis, thus we combined coffee with caffeine (coffee/caffeine) to increase the statistical power in subgroup analysis. 20 studies [14,15,17–25,31–34,38,41,43,46,47] were

case-control designs and the remaining 17 studies were cohort designs. 3 studies [22,23,25] examined the association of coffee intake with breast cancer among BRCA1 mutation carriers.

Quantitative synthesis

The results are summarized in Table 2.

Overall association of coffee and caffeine with breast cancer risk

Compared to the lowest category, the combined RR (95% CI) of breast cancer for the highest category was 0.99 (0.94–1.04) for caffeine, 0.97 (0.93–1.00) for coffee (Fig. 2) and 0.97 (0.94–1.01) for coffee/caffeine, respectively. 12 studies [6,10,13,18–20,23,32,41,43,45,47] provided the result for decaffeinated coffee, and the RR (95% CI) of breast cancer was 0.98 (0.92–1.05) for the highest category versus the lowest category of decaffeinated coffee. The age-adjusted RR (95% CI) of breast cancer was 0.96 (0.92–1.01) for coffee/caffeine [7,13,15,16,24,32,35,37,44,45]. Moderate or lower between-study heterogeneity ($I^2 < 50\%$) was found in all analyses.

Subgroup analysis by study design

An inverse but not significant association was found for both cohort studies (RR = 0.98, 95%CI = 0.95–1.02) and case-control studies (RR = 0.94, 95%CI = 0.89–1.00). For cohort studies, similar results were found between studies with the follow-up duration > 10 years (RR = 0.99, 95%CI = 0.94–1.04) [6–8,10–12,16,35,36,39,40,45] and < 10 years (RR = 0.98, 95%CI = 0.91–1.05) [9,13,37,42,44]. For case-control studies, the association was still not significant for population-based case-control studies (RR = 0.92, 95%CI = 0.84–1.01) [14,15,18,19,21,31,33,34,38,41] and hospital-based case-control studies (RR = 0.96, 95%CI = 0.89–1.05) [17,20,22–25,32,43,46,47], respectively. Moderate or lower between-study heterogeneity ($I^2 < 50\%$) was found in all analyses.

Subgroup analysis by population subgroups

The negative association of coffee/caffeine intake with breast cancer was consistent across different population subgroups by location where the study was conducted (America [6,8,10,13,16,18,19,32,37,41,43,45], Europe [7,9,11,12,15,17,20–22,34–36,38,40,44,46,47], or Asia [24,31,39,42]), BMI (< 25 kg/m² [6,9,40,45] or > 25 kg/m² [6,9,40,45]), ER and PR status (ER + PR + [6,7,10,12,13,45], ER + PR – [6,7,12,13], ER – PR + [12,13] or ER – PR – [6,7,10,12,13,45]), and breast cancer stage (invasive [6–9,12,13,15,40,45] or in situ [13]). In the subgroup analysis by menopausal status, the association was not significant for premenopausal women [6,8,10,14,21,33,38,40,43,45], but a significant association was found for postmenopausal women (RR = 0.94, 95%CI = 0.8–0.99, $P = 0.02$) [6,10,13–15,21,33,37,40,43,45]. A strong and significant association was found between coffee intake and breast cancer risk among BRCA1 mutation carriers (RR = 0.69, 95%CI = 0.53–0.89, $P = 0.005$). Moderate or lower between-study heterogeneity ($I^2 < 50\%$) was found in all analyses.

Subgroup analysis by adjustment for covariates

No significant association was found in stratified analysis by adjustment (yes or no) for the following covariates: smoking and/or alcohol, BMI, energy intake, physical activity, oral contraceptive use, postmenopausal hormone replacement therapy use, family history of breast cancer and history of benign breast disease. Moderate or lower between-study heterogeneity ($I^2 < 50\%$) was found in all analyses.

Dose-response analysis

Association of coffee intake with breast cancer

Overall, data from 20 studies [6,7,9–15,23,24,32,34,35,37,40,41,43–45] including 41,805 breast cancer cases were used. Linear relationship was found between coffee intake and breast cancer

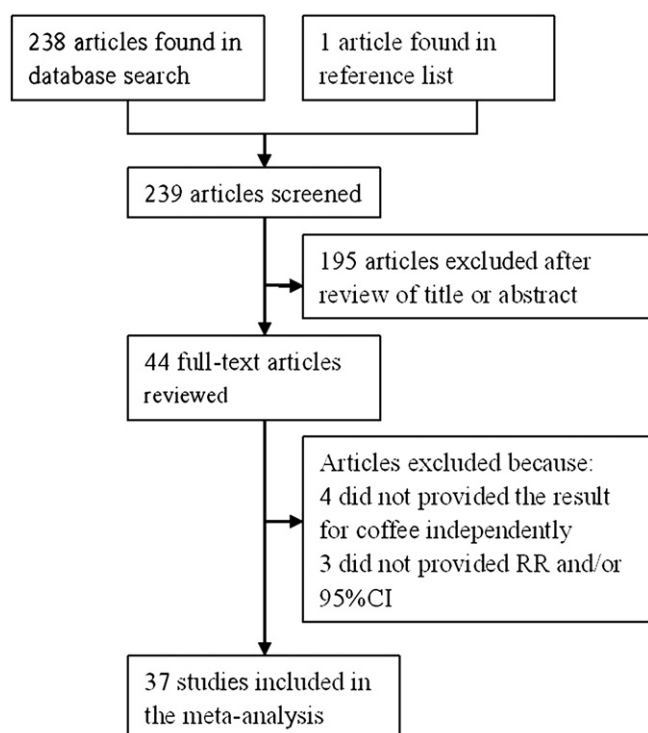


Fig. 1. Selection of studies for inclusion in this meta-analysis.

Table 1
Characteristics of studies included in the meta-analysis on coffee and caffeine intake and breast cancer.

Author, Year [Ref.]	Study design ^a , age (years)	Study name Follow-up (years)	Country	Sample size (cases)	Exposure	RR (95%CI) for highest vs. lowest category	Adjustment for covariates ^b
Snowdon et al. 1984 [16]	Cohort, >40	None, 21	USA	23,912 (176)	Coffee	>2 vs. <1 cups/day 0.9 (0.6–1.3)	Age and sex
Lubin et al. 1985 [31]	CC, unclear	–	Israel	1431 (813)	Coffee	≥4 vs. 0 cup/day 0.6 (0.2–0.9)	None
Rosenberg et al. 1985 [32]	CC, 30–69	–	USA	4152 (2651)	Coffee	>5 vs. 0 cup/day 1.2 (0.9–1.6)	Age, race, religion, smoking, age at menarche, age at first pregnancy, parity, menopause status, age at menopause, FHBC, BMI, education, tea, alcohol, location of the hospital, year of interview, number of previous non-obstetric hospitalizations
Katsouyanni et al. 1986 [17]	CC, 54.7/53.7	–	Greece	240 (120)	Coffee	Consumption level (3th vs. 1th) 1.12 (0.48–2.59)	None
La Vecchia et al. 1986 [46]	CC, unclear	–	Italy	1232 (616)	Coffee	≥4 vs. 0 cup/day 1.1 (0.7–1.7)	Age, geographic area, parity, age at first birth, age at menarche and menopause, OC and HRT use, smoking and alcohol
Schairer et al. 1987 [18]	CC, unclear	BCDDP	USA	3392 (1510)	Coffee	≥5 vs. 0 cup/day 1.0 (0.8–1.3)	None
Rohan et al. 1988 [33]	CC, 20–74	–	Australia	902 (451)	Caffeine	390.0 vs. 158.2 mg/day 1.20 (0.79–1.83)	None
Ewertz et al. 1990 [34]	CC, unclear	–	Denmark	2822 (1486)	Coffee	>10 vs. 0 cup/day 0.81 (0.57–1.15)	Age at diagnosis and place of residence
Vatten et al. 1990 [35]	Cohort, 35–51	None, 12	Norway	14,593 (152)	Coffee	≥5 vs. ≤2 cups/day 0.8 (0.5–1.4)	Age
Hoyer et al. 1992 [36]	Cohort, 30–80	Glostrup Population Studies, 26	Denmark	5207 (51)	Coffee	≥7 vs. ≤2 cups/day 1.7 (0.7–4.3)	Social class, age at menarche, menopause status, number of full-term pregnancies, height, weight, BMI, alcohol and smoking
McLaughlin et al. 1992 [19]	CC, 20–79	–	USA	3234 (1617)	Coffee	Drinker vs. nondrinker 0.98 (0.76–1.26)	Age, county of residence, race, menstrual status, age at first live birth, HBBD, FHBC and alcohol
Folsom et al. 1993 [37]	Cohort, 55–69	Iowa Women's Health Study, 5	USA	34,388 (580)	Coffee	≥4 vs. 0 cup/day 1.02 (0.79–1.30)	Age, waist/hip ratio, number of live births, age at first live birth, age at menarche, FHBC, Family history × waist/hip ratio, and family history × number of live births
Levi et al. 1993 [20]	CC, 30–75	–	Switzerland	425 (107)	Coffee	≥503.8 vs. ≤65.2 mg/day 1.02 (0.78–1.33)	None
Smith et al. 1994 [38]	CC, <36	–	UK	1510 (755)	Caffeine	Consumption level (3th vs. 1th) 0.98 (0.53–1.79)	Age at menarche, nulliparity, age at first full-term pregnancy, breastfeeding, FHBC, OC use, HBBD, alcohol at 18 and smoking
Tavani et al. 1998 [47]	CC, <75	–	Italy	11,488 (5984)	Coffee	>301 vs. <100 mg/day 0.83 (0.59–1.17)	Study/center, age, education, BMI, smoking, alcohol, age at menarche and menopause, parity and age at first birth, OC use, HRT, HBBD, FHBC
Key et al. 1999 [39]	Cohort, <40– >80	RERFLS, >14	Japan	34,759 (427)	Coffee	≥4 vs. <2 cups/day 0.96 (0.83–1.11)	Age, calendar period, city, age at time of bombing and radiation dose
Mannisto et al. 1999 [21]	CC, 25–75	–	Finland	764 (310)	Coffee	≥5 vs. ≤1 times/day 1.19 (0.93–1.52)	Age, area (rural/urban), age at menarche, age at first full-term pregnancy, OC use, HRT, FHBC, HBBD, education, alcohol, smoking, leisure activity and waist/hip ratio.
Michels et al. 2002 [40]	Cohort, 40–76	Swedish Mammography Cohort, 11	Sweden	59,036 (1271)	Caffeine	Consumption level (4th vs. 1th) Premenopausal 1.8 (0.8–4.3) Postmenopausal 0.5 (0.3–1.0)	Age, family history of breast cancer, height, BMI, education, parity, age at first birth, alcohol and caloric intake
Wu et al. 2003 [41]	CC, 25–74	–	USA	1095 (501)	Coffee	308.8 vs. 83.8 g/day 1.04 (0.87–1.24)	Age, Asian ethnicity, birthplace, education, age at menarche, pregnancy, current BMI, caloric intake,
						>240 vs. 0 ml/day 0.77 (0.53–1.12)	

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Table 1 (continued)

Author, Year [Ref.]	Study design ^a , age (years)	Study name Follow-up (years)	Country	Sample size (cases)	Exposure	RR (95%CI) for highest vs. lowest category	Adjustment for covariates ^b
Suzuki et al. 2004 [42]	Cohort, 40–64	None, 8	Japan	35,004 (222)	Coffee	≥ 1 vs. 0 cup/day 0.81 (0.55–1.18)	menopausal status, HRT use, intake of soy and dark green vegetables, smoking, alcohol, physical activity, FHBR
Baker et al. 2006 [43]	CC, 21–94	–	USA	3827 (1932)	Coffee	> 4 vs. 0 cup/day Premenopausal 0.62 (0.39–0.98) Postmenopausal 0.99 (0.79–1.23)	Age, types of health insurance, age at menarche, menopausal status, age at first birth, parity, FHBC, smoking, alcohol, BMI and tea
Gronwald et al. 2006 [22]	CC, 43.9/36.4	–	Poland	696 (348)	Coffee	Drinker vs. nondrinker 0.8 (0.5–1.1)	Age, residence, and age at birth of first child
Hivonen et al. 2006 [44]	Cohort, 35–60	SU.VI.MAX Study, 6.6	Finland	4396 (95)	Coffee	≥ 253 vs. < 111 ml/day 1.10 (0.66–1.84)	Age and residence
Nkondjock et al. 2006 [23]	CC, < 64	–	Four countries	1690 (845)	Coffee	≥ 6 vs. 0 cup/day 0.51 (0.26–0.98)	Year of birth, age at diagnosis, age at menarche, parity, smoking, breast-feeding and OC use
Hirose et al. 2007 [24]	CC, 40–79	HERPACC study	Japan	14,547 (2122)	Coffee Caffeine	> 3 vs. 0 cup/day 1.04 (0.85–1.28) > 270 vs. ≤ 120 mg/day 1.03 (0.89–1.20)	Age, smoking, number of children, OC use, FHBC and menopausal status
Kotsopoulos et al. 2007 [25]	CC, 52.4/43.1	–	Canada	411 (170)	Coffee	Drinker vs. nondrinker 0.61 (0.38–0.97)	Parity, smoking, OC use, alcohol and BMI at age 30
Ganmaa et al. 2008 [45]	Cohort, 30–55	Nurses' Health Study, 22	USA	85,987 (5272)	Coffee Caffeine	≥ 6 cups/day vs. < 1 cup/month 0.92 (0.82–1.03) 816 vs. 51 mg/day 0.93 (0.85–1.01)	Age, year, motivation for consultation, parity, age at first delivery, smoking, alcohol, type of breakfast, fondness of salty and fatty foods, fruit, vegetable, beef, fish, carrot, exercise and BMI
Ishitani et al. 2008 [6]	Cohort, > 45	Women's Health Study, 10	USA	38,432 (1188)	Coffee Caffeine	≥ 4 vs. 0 cup/day 1.08 (0.89–1.30) > 486.3 vs. ≤ 68.0 mg/day 1.02 (0.84–1.22)	Year of birth, age at menarche, parity and smoking
Bissonauth et al. 2009 [14]	CC, < 35– > 65	–	Canada	560 (280)	Coffee	> 8 vs. ≤ 2 cups/day 1.4 (1.09–2.24)	Age, smoking, BMI, physical activity, height, alcohol, FHBC, HBBD, menopausal status, age at menopause, HRT use, age at menarche, weight change after 18 and duration of HRT
Larsson et al. 2009 [7]	Cohort, 40–76	Swedish Mammography Cohort, 17.4	Sweden	61,433 (2952)	Coffee	≥ 4 vs. < 1 cups/day 1.02 (0.87–1.20)	Age, randomized treatment assignment, BMI, physical activity, energy, alcohol, multivitamin use, age at menopause, age at menarche, age at first pregnancy lasting ≥ 6 months, number of pregnancies lasting ≥ 6 months, menopausal status, HRT use, prior hysterectomy, prior bilateral oophorectomy, smoking, FHBC and HBBD
Wilson et al. 2009 [8]	Cohort, 25–42	Nurses' Health Study II, 14	USA	90,628 (1179)	Coffee	consumption level (5th vs. 1th) 0.92 (0.77–1.11)	Age, education, physical activity, smoking, coffee and energy
Bhoo et al. 2010 [9]	Cohort, 20–70	EPIC-NL Cohort, 9.6	Netherlands	27,323 (681)	Coffee	> 5 vs. 0 cup/day 0.94 (0.72–1.24)	Age, education, BMI, height, parity, age at first birth, age at menarche, age at menopause, OC use, HRT use, FHBC, energy, alcohol and tea
Boggs et al. 2010 [10]	Cohort, 21–69	Black Women's Health Study, 12	USA	52,062 (1268)	Coffee Caffeine	≥ 4 vs. < 1 cups/day 1.03 (0.77–1.39) ≥ 209 vs. < 16 mg/day 1.04 (0.87–1.24)	BMI, height, OC use, parity and age at first birth, age at menarche, FHBC, HBBD, smoking, physical activity, animal fat, glycemic load, alcohol and energy

Nilsson et al. 2010 [11]	Cohort, >30	Vasterbotten Intervention Project, 15	Sweden	64,603 (3034)	Coffee	≥4 vs. <1 cups/day 0.92 (0.68–1.25)	Age, sex, BMI, smoking, education, and recreational physical activity
Fagherazzi et al. 2011 [12]	Cohort, 40–65	E3N cohort, 11	France	67,703 (2868)	Coffee Caffeine	>3 vs. 0 cup/day 1.02 (0.90–1.16)	Age, energy, OC use, age at menarche, age at menopause, number of children, age at first pregnancy, FHBC, education, HRT use, HBBB, menopausal status and BMI
Li et al. 2011 [15]	CC, 50–74	– MARIE	Sweden Germany	5929 (2818) 8046 (2651)	Coffee Coffee	>5 vs. ≤1 cups/day 0.84 (0.66–1.06) >5 vs. ≤1 cups/day 0.87 (0.71–1.07)	Age, HRT use, smoking, education and alcohol
Gierach et al. 2012 [13]	Cohort, 50–71	Health-AARP Diet and Health Study cohort, 5,2/9,8	USA	198,404 (9915)	Coffee	Coffee: >4 vs. 0 cup/day 0.98 (0.91–1.07)	Age, race/ethnicity, education, BMI, smoking, alcohol, proportion of total energy from fat, age at first live birth, HRT use, history of breast biopsy and FHBC

^a CC: case-control study.

^b BMI: body mass index, OC: oral contraceptive, HRT: hormone replacement therapy, FHBC: family history of breast cancer, HBBB: history of benign breast disease.

risk (P for nonlinearity = 0.49), and the RR (95% CI) of breast cancer was 1.00 (0.98–1.01), 0.99 (0.96–1.02), 0.98 (0.95–1.01), 0.97 (0.94–1.01), 0.95 (0.92–1.00), 0.94 (0.90–0.99), 0.93 (0.88–0.98) and 0.92 (0.85–0.98) for 1, 2, 3, 4, 5, 6, 7 and 8 cups/day of coffee intake, respectively. The risk of breast cancer decreased by 2% (RR = 0.98, 95%CI = 0.96–1.00, P = 0.053) for every 2 cups/day increment in coffee intake.

For case-control study, data from 8 studies [14,15,23,24,32,34,41,43] including 15,195 breast cancer cases were used. Linear relationship was found between coffee intake and breast cancer risk (P for nonlinearity = 0.18), and the RR (95% CI) of breast cancer was 1.01 (0.98–1.04), 1.01 (0.95–1.07), 1.00 (0.94–1.06), 0.97 (0.92–1.04), 0.95 (0.89–1.01), 0.93 (0.86–1.00), 0.90 (0.82–0.99) and 0.88 (0.79–0.99) for 1, 2, 3, 4, 5, 6, 7 and 8 cups/day of coffee intake, respectively. The risk of breast cancer decreased by 3% (RR = 0.97, 95%CI = 0.93–1.02, P = 0.21) for every 2 cups/day increment in coffee intake.

For cohort study, data from 12 studies including 26,610 breast cancer cases were used. Linear relationship was found between coffee intake and breast cancer risk (P for nonlinearity = 0.72), and the RR (95% CI) of breast cancer was 0.99 (0.97–1.01), 0.98 (0.94–1.02), 0.97 (0.94–1.01), 0.97 (0.93–1.00), 0.96 (0.92–1.01), 0.96 (0.91–1.01) and 0.95 (0.89–1.02) for 1, 2, 3, 4, 5, 6 and 7 cups/day of coffee intake, respectively (Fig. 3). The risk of breast cancer decreased by 2% (RR = 0.98, 95%CI = 0.97–1.00, P = 0.08) for every 2 cups/day increment in coffee intake.

Association of caffeine intake with breast cancer

Data from 7 studies [6,10,12,24,37,38,45] including 14,020 breast cancer cases were used. Linear relationship was found between caffeine intake and breast cancer risk (P for nonlinearity = 0.61), and the RR (95% CI) of breast cancer was 1.00 (0.98–1.03), 1.00 (0.96–1.05), 1.00 (0.95–1.04), 0.99 (0.95–1.03), 0.98 (0.94–1.03), 0.98 (0.93–1.03), 0.97 (0.92–1.03) and 0.96 (0.90–1.03) for 100, 200, 300, 400, 500, 600, 700 and 800 mg/day of caffeine intake, respectively (Fig. 4). The risk of breast cancer decreased by 1% (RR = 0.99, 95%CI = 0.98–1.01, P = 0.52) for every 200 mg/day increment in caffeine intake. Considering the limited number of studies included, stratified analysis by study design (case-control and cohort) was not performed.

Sensitivity analysis and publication bias

Sensitivity analysis showed that no individual study had excessive influence on the pooled effect between breast risk and coffee and caffeine intake, respectively. Egger test showed no evidence of significant publication bias for the analysis between breast cancer risk and coffee (P = 0.23) and caffeine (P = 0.35), respectively.

Discussion

Findings from this meta-analysis suggested that coffee/caffeine might be weakly associated with breast cancer risk for postmenopausal women, and a strong and significant association of coffee with breast cancer risk was found among BRCA1 mutation carriers. A linear but not significant dose-response relationship was found between breast cancer risk and coffee and caffeine intake, respectively.

Coffee is a complex chemical mixture that contains many compounds including caffeine, acrylamide, various polyphenols, etc. which can play a dual role as both a carcinogen and a chemo-preventive agent. Animal tests showed that caffeine can both promote and suppress mammary tumors [48]. And acrylamide was also shown to increase breast cancer risk [8]. Experimental studies suggested that polyphenols, like chlorogenic acid and caffeic acid and lignans could interfere at the initiation, promotion and progression of cancer [49]. Furthermore, previous studies suggested that coffee and caffeine were inversely correlated with sex hormones (testosterone and estradiol)

Table 2

Pooled measures on the relation of coffee and caffeine to breast cancer.

	Fixed effect model (FEM)	Random effect model (REM)	P^a	I^2 (%)	Number of studies	Number of breast cancer cases
Caffeine	0.99 (0.94–1.04)	0.99 (0.94–1.04)	0.73	0.00	9	15,775
Coffee	0.97 (0.93–1.00)	0.97 (0.92–1.01)	0.09	14.2	34	56,541
Decaffeinated coffee	0.98 (0.92–1.05)	0.97 (0.89–1.06)	0.55	29.7	12	31,790
Caffeine/coffee	0.97 (0.94–1.01)	0.97 (0.93–1.01)	0.09	12.3	37	59,018
Age-adjusted RR	0.96 (0.92–1.01)	0.95 (0.88–1.03)	0.15	42.6	10	29,384
BRCA1 mutation carriers	0.69 (0.53–0.89)	0.69 (0.53–0.89)	0.01	0.00	3	1363
Study design						
Cohort study	0.98 (0.95–1.02)	0.98 (0.95–1.02)	0.44	0.00	17	30,931
Case-control study	0.94 (0.89–1.00)	0.93 (0.85–1.01)	0.06	34.0	20	28,087
Follow-up duration for cohort studies						
> 10 years	0.99 (0.94–1.04)	0.99 (0.94–1.04)	0.62	0.00	12	19,438
< 10 years	0.98 (0.91–1.05)	0.98 (0.91–1.05)	0.50	0.00	5	11,493
Source of controls for case-control study						
Population-based	0.92 (0.84–1.01)	0.92 (0.81–1.04)	0.09	37.5	10	13,192
Hospital-based	0.96 (0.89–1.05)	0.94 (0.84–1.06)	0.13	33.7	10	14,895
Menopausal status						
Premenopausal	0.97 (0.87–1.07)	0.97 (0.86–1.09)	0.50	10.1	10	4842
Postmenopausal	0.94 (0.89–0.99)	0.94 (0.89–1.00)	0.02	9.10	11	24,188
Country						
USA	0.97 (0.92–1.01)	0.97 (0.92–1.01)	0.14	0.00	12	27,789
Asia	1.03 (0.91–1.16)	1.00 (0.83–1.21)	0.66	37.9	4	3184
Europe	0.97 (0.92–1.02)	0.97 (0.92–1.02)	0.24	0.00	17	26,299
Body mass index (BMI)						
< 25 kg/m ²	1.01 (0.90–1.12)	1.01 (0.90–1.12)	0.91	0.00	4	4372
> 25 kg/m ²	0.92 (0.82–1.04)	0.92 (0.82–1.04)	0.18	0.00	4	4029
Estrogen receptor (ER) and progesterone receptor (PR) status (positive: +, negative: –)						
ER + PR +	0.96 (0.89–1.04)	0.96 (0.88–1.06)	0.28	16.1	6	8094
ER + PR–	1.05 (0.88–1.26)	1.05 (0.88–1.26)	0.59	0.00	4	1449
ER–PR +	0.70 (0.43–1.15)	0.70 (0.43–1.15)	0.16	0.00	2	155
ER–PR–	0.95 (0.83–1.09)	0.97 (0.83–1.13)	0.48	15.8	6	2214
Breast cancer stage						
Invasive	0.97 (0.92–1.01)	0.97 (0.92–1.01)	0.15	0.00	9	27,828
In situ	1.02 (0.85–1.24)	1.02 (0.85–1.24)	–	–	1	1892
Adjustment for smoking and/or alcohol						
No	1.00 (0.93–1.07)	1.00 (0.93–1.07)	0.98	0.00	12	10,222
Yes	0.96 (0.93–1.00)	0.96 (0.92–1.01)	0.06	20.0	25	48,796
Adjustment for BMI						
No	0.95 (0.88–1.01)	0.94 (0.86–1.03)	0.12	26.0	22	17,065
Yes	0.98 (0.94–1.02)	0.98 (0.94–1.02)	0.29	0.00	15	41,953
Adjustment for energy intake						
No	0.94 (0.90–0.99)	0.94 (0.89–1.00)	0.02	16.1	27	36,915
Yes	1.00 (0.96–1.05)	1.00 (0.96–1.05)	0.97	0.00	10	22,103
Adjustment for physical activity						
No	0.98 (0.94–1.02)	0.97 (0.93–1.02)	0.24	4.90	27	43,183
Yes	0.97 (0.91–1.02)	0.97 (0.90–1.06)	0.22	33.1	10	15,835
Adjustment for oral contraceptive (OC) use						
No	0.97 (0.93–1.01)	0.97 (0.92–1.02)	0.12	11.9	25	41,117
Yes	0.98 (0.93–1.04)	0.97 (0.90–1.04)	0.50	19.2	12	17,901
Adjustment for postmenopausal hormone replacement therapy (HRT) ^b						
No	0.99 (0.93–1.05)	0.98 (0.91–1.05)	0.75	16.6	24	36,343
Yes	0.97 (0.93–1.01)	0.97 (0.92–1.02)	0.13	15.3	11	20,740
Adjustment for family history of breast cancer (FHBC)						
No	0.97 (0.91–1.03)	0.95 (0.88–1.03)	0.27	26.6	20	20,980
Yes	0.98 (0.94–1.01)	0.98 (0.94–1.01)	0.20	0.00	17	38,038
Adjustment for history of benign breast disease (HBBD)						
No	0.97 (0.94–1.02)	0.98 (0.93–1.03)	0.34	12.3	29	39,845
Yes	0.96 (0.91–1.01)	0.96 (0.90–1.02)	0.14	20.0	8	19,173

^a P value for significance test of RR in FEM because all $I^2 < 50\%$.^b 2 studies with premenopausal women only were excluded in this analysis.

[50,51] and positively associated with sex hormone-binding globulin (SHBG) levels among postmenopausal women [50,52]. And increased risks of breast cancer were found associated with elevated serum concentrations of testosterone and estradiol among postmenopausal women [53–55], while high levels of SHBG was found associated with decreased risk of breast cancer among postmenopausal women [53]. This might partially explain the observed association among postmenopausal women in this meta-analysis. BRCA1 is a well-identified gene involved in breast cancer, and the cumulative risk of breast cancer

was estimated to be 54% by age 60 years, and 85% by age 70 among BRCA1 mutation carriers [56]. Thus exploring the effect of coffee on breast cancer among BRCA1 mutation carriers also received much attention [22,23,25]. However, the mechanism for BRCA1 mutation interacting with coffee/caffeine for breast cancer risk is still unclear. The previous study indicated that majority of BRCA1-associated breast cancers are estrogen-receptor negative (ER–) [57], and protection was also somewhat stronger for ER– breast cancer in this meta-analysis, especially for ER–PR+ breast cancer (RR = 0.70). However,

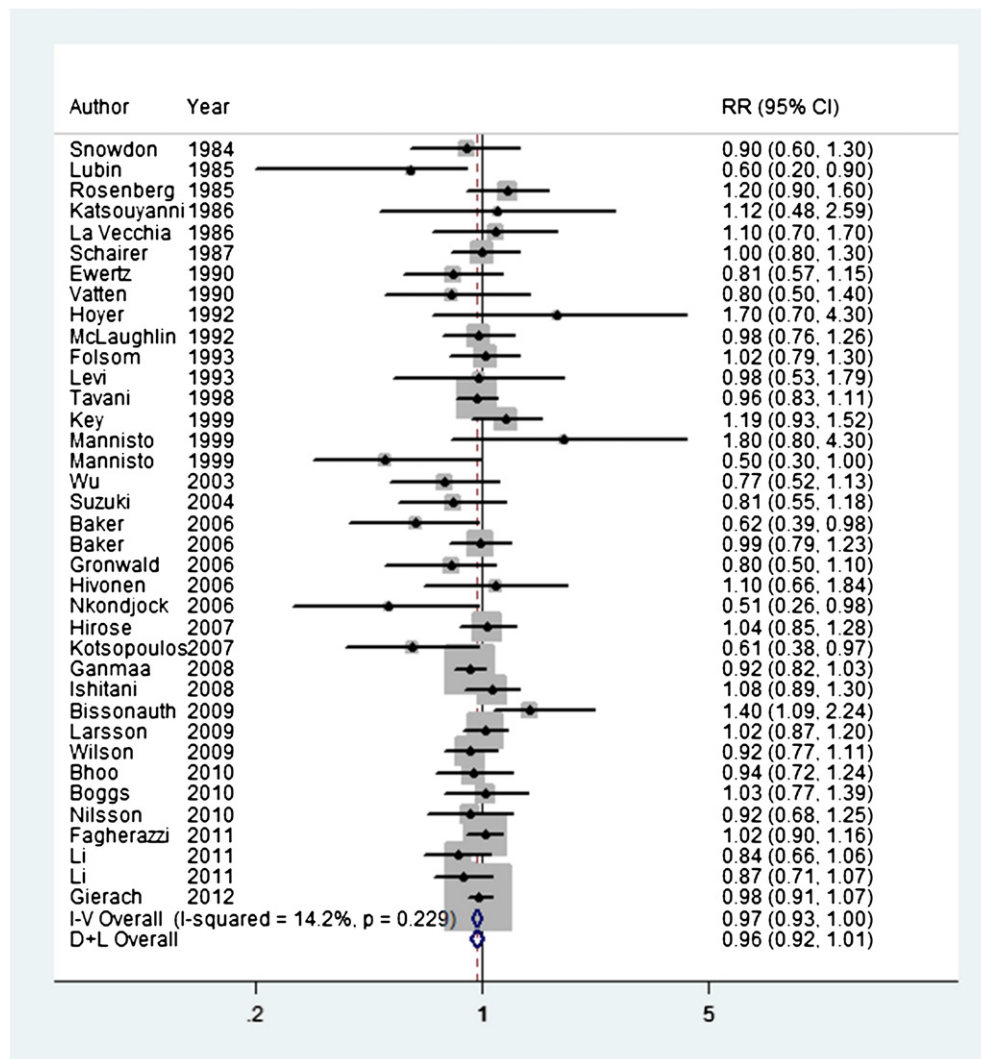


Fig. 2. The multivariate-adjusted risk of breast cancer for the highest vs. lowest categories of coffee intake. The size of gray box is positively proportional to the weight assigned to each study, which is inversely proportional to the standard error of the RR, and horizontal lines represent the 95% confidence intervals. D + L denotes random effect model (REM), I-V denotes fixed effect model (FEM).

only 2 results were included in this meta-analysis on coffee/caffeine and ER – PR + breast cancer, which needs to be confirmed in the further studies.

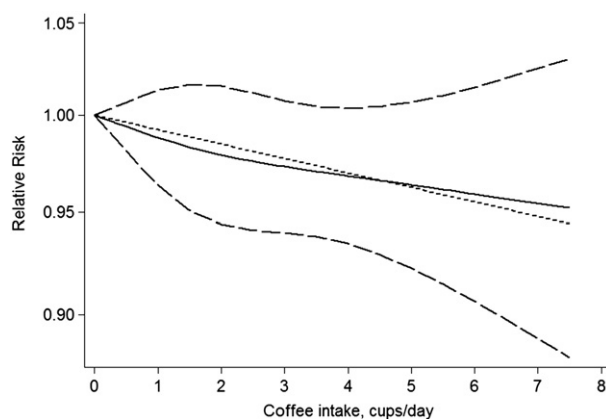


Fig. 3. The dose-response analysis between coffee intake and breast cancer risk in cohort studies with restricted cubic splines in a multivariate random-effects dose-response model. The solid line and the long dash line represent the estimated relative risk and its 95% confidence interval. Short dash line represents the linear relationship.

Observational studies cannot prove causality. However, results of our meta-analysis do not support an obvious causation overall according to the Hill criteria [58]. (1) Strength: the magnitude of this association is negligible overall; (2) Consistency: negative association was found in almost all stratified analyses and low between-study heterogeneity was found in most subgroup analysis; (3) Temporality: negative association from the prospective studies does not consist of an appropriate temporal relationship, in which the exposure precedes breast cancer incidence; (4) Plausibility and coherence: coffee and caffeine can both promote and suppress mammary tumors. (5) Biological gradient: the linear but not significant dose-response relationship does not meet an obviously biological gradient. Although marginally significant association was found for coffee consumers of ≥ 6 cups/day overall, it has little public health significance, considering the fact that Finnish who consume the most amount of coffee (12 kg per person) in the world do not consume that much of coffee (Current Worldwide Annual Coffee Consumption per capita around 2009, <http://chartsbin.com/view/581>, accessed 1/10/2013).

A major strength of this study was the large number of participants included, allowing a much greater possibility of reaching reasonable conclusions and conducting subgroup analysis. Dose-response analysis was also performed to better describe the association of breast cancer risk with coffee and caffeine intake. However, there were some limitations in this meta-analysis. First, only 3 studies [22,23,25] were included

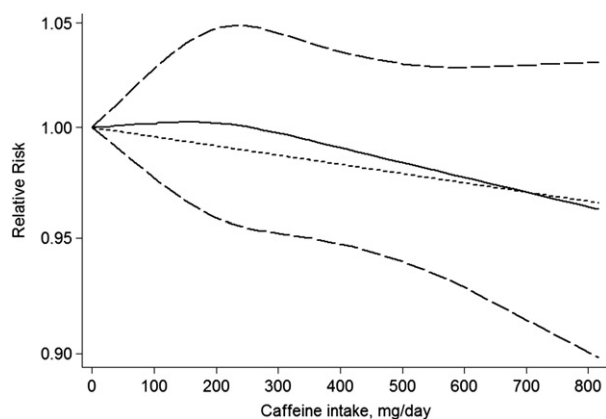


Fig. 4. The dose–response analysis between caffeine intake and breast cancer risk with restricted cubic splines in a multivariate random-effects dose–response model. The solid line and the long dash line represent the estimated relative risk and its 95% confidence interval. Short dash line represents the linear relationship.

for BRCA1 mutation carriers, thus the observed association needs to be confirmed in the further studies. Second, misclassification of coffee consumption was inevitable in the original studies. However, results from validation studies suggested that coffee consumption was assessed with relatively high validity [59]. Third, although we extracted the RRs that reflected the greatest degree of control for potential confounders, but the extent to which they were adjusted varied in the original studies. However, the strength of effect estimates was similar between age-adjusted RR and multivariate-adjusted RR, and similar result was also found by adjustment (yes or no) of selected key covariates. Finally, in a meta-analysis of published studies, it is possible that an observed effect might suffer from publication bias because studies with null results tend not to be published. However, no significant publication bias was detected in this meta-analysis.

In summary, results from this meta-analysis indicated that coffee/caffeine might be weakly associated with breast cancer risk for postmenopausal women, and the association of coffee with breast cancer risk for BRCA1 mutation carriers deserves further investigation.

Conflict of interest statement

None.

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